# Server Management and Maintenance

## Objectives:

At the end of this episode, I will be able to:

1. Identify and determine appropriate management tools and technologies for maintaining servers.

Exam Objective: *1.3 - Given a scenario, perform server hardware maintenance.*

Description: In this episode, the viewer will identify local administrative interfaces and remote management techniques use to connect to, configure, manage, and maintain servers.

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- Local hardware administration
  - Keyboard-video-mouse (KVM) (Lucid Chart)
  - Console connections (Lucid Chart)
  - Serial connectivity
  - Remote Desktop Connections
  - Virtual administration console
    - Hyper-V Manager
    - Windows Admin
    - vSphere
  - Hot swappable components
    - on-prem administration and maintenance
    - Drives, fans, power supplies
- Out-of-band management (Lucid Chart)
  - Dell's integrated Remote Access Controller (iDRAC)
  - HPE's integrated Lights Out (iLO)
    - Remote console access
    - Remote power on/off
    - Remote monitoring
  - Internet protocol keyboard-video-mouse (IP KVM)
  - Firmware management

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## External Resources:

During this episode, you can reference the following external resources for supplementary tools and information:

- Virtual Administration Consoles
  - vSphere - https://www.vmware.com/products/vsphere.html
  - Windows Admin - https://docs.microsoft.com/en-us/windows-server/manage/windows-admin-center/use/manage-virtual-machines
- Out-of-band Management
  - HPE iLO - https://www.hpe.com/us/en/servers/integrated-lights-out-ilo.html
  - Dell iDRAC - https://www.delltechnologies.com/en-us/solutions/openmanage/idrac.htm